6 Radiative transfer: Blackbody radiation, energy balance, idealised radiative-convective equilibrium

Reading for Day 6: Pierrehumbert Principles of Planetary Climate, Section 3.1-3.3.

6.1 Description of radiation

6.1.1 Solid angle

Let us consider a ray (i.e., a unit vector pointing in a given direction). This ray can be described by spherical coordinates (θ, ϕ), and contains information about direction but not length. A solid angle Ω is a measure of a set of rays pointing in a range of directions. For example, if we integrate over all directions, $\Omega = 4\pi$, and if we integrate over all upward directions $\Omega = 2\pi$. Figure 6.1 shows the difference between solid angle and the more typical plane angle.

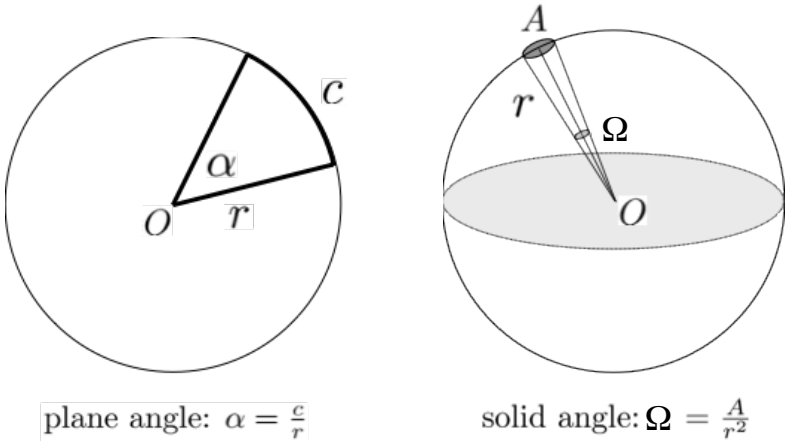


Figure 6.1: Difference between a planar angle (left) and solid angle (right). Figure courtesy Ray Pierrehumbert.

Figure 6.2 shows the differential solid angle, $d\Omega = \sin\theta d\theta d\phi$. We can also express $d\Omega = -d\mu d\phi$, where $\mu \equiv \cos\theta$. Thus, the projected solid angle $d\Omega_\perp = \cos\zeta d\Omega$, where ζ is the *zenith angle* (i.e., the angle between the direction of a ray and the normal to the surface that the ray intersects).

6.1.2 Radiance

Radiance, typically written as I_ν (or in the case of blackbody radiance B_ν) is the rate at which radiation energy flows per unit frequency (or wavelength) per unit solid angle per unit time per unit area of a surface perpendicular to the direction identified by the vector $\vec{\mathcal{L}}$ in Figure 6.2. Radiance is sometimes also called "intensity." The units of radiance are W m^{-2} Hz^{-1} steradian^{-1} (replace Hz with length, e.g., μm if using the per wavelength form). Radiance is positive by definition. Thus, $I_\nu d\nu d\Omega dA$ is the radiation energy between frequency ν and $\nu + d\nu$ and directions within a solid angle $d\Omega$ of the vector $\vec{\mathcal{L}}$ that flows per unit time through the area dA perpendicular to the vector $\vec{\mathcal{L}}$.

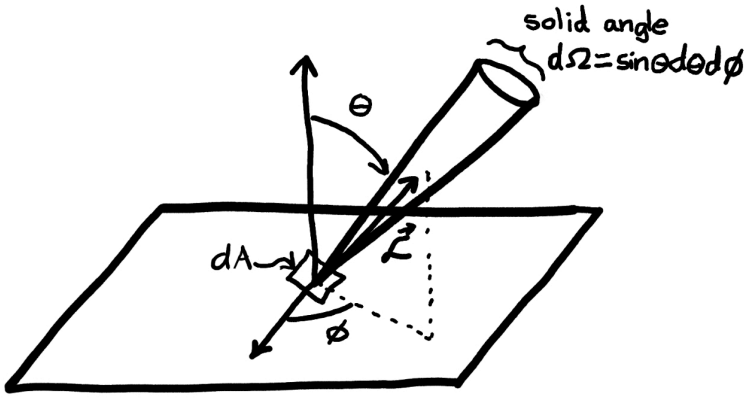


Figure 6.2: Schematic showing radiance through a surface perpendicular to some direction identified by the vector $\vec{\mathcal{L}}$. Figure courtesy Adam Showman.

6.1.3 Flux

Flux (typically denoted F_ν) is the total radiation energy per unit frequency (or wavelength) per unit area, of all directions, that passes through some surface. Thus, we can relate flux to radiance as

$$F_\nu = \int_{4\pi} I_\nu \cos\theta d\Omega = \int_0^{2\pi} \int_0^{\pi} I_\nu(\phi,\theta) \cos\theta \sin\theta d\theta d\phi. \tag{6.1}$$

We can further consider upward and downward fluxes when integrating only over thee upward or downward directions:

$$F_\nu^+ = \int_0^{2\pi} \int_0^{\pi/2} I_\nu(\phi,\theta) \cos\theta \sin\theta d\theta d\phi, \tag{6.2}$$

$$F_\nu^- = -\int_0^{2\pi} \int_{\pi/2}^{\pi} I_\nu(\phi,\theta) \cos\theta \sin\theta d\theta d\phi. \tag{6.3}$$

We have written this such that F_ν^+ and F_ν^- are both positive and $F_\nu = F_\nu^+ - F_\nu^-$. The total flux is the the integral over frequency or wavelength,

$$F \equiv \int_0^\infty F_\nu d\nu = \int_0^\infty F_\lambda d\lambda. \tag{6.4}$$

Note that for *isotropic* radiation, $F_\nu^+ = F_\nu^- = \pi I_\nu$, and thus $F_\nu = 0$.

6.2 Blackbody radiance

A detailed understanding of planetary climate requires an understanding of the way light is transferred, absorbed, and scattered through a planetary atmosphere. This requires a full wavelength-dependent understanding of these processes, which is necessarily complicated. However, we can first gain understanding by using a simpler class of models that consider only the bulk energy balance of the planet, assuming that emission follows that of a blackbody (i.e., the radiation field inside a fixed-temperature enclosed region in thermodynamic equilibrium).

The radiance of a blackbody depends only on temperature and frequency (or wavelength), and is given by the Planck function

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT-1}} = \frac{2k_B^3T^3}{h^2c^2}\frac{u^3}{e^u-1}, \tag{6.5}$$

where h is Planck's constant, c is the speed of light in a vacuum, k_B is Boltzmann's constant, T is temperature, and ν is the frequency of radiation. u is a dimensionless quantity equal to half the ratio of the photon energy to the thermal energy in one degree of freedom of matter in thermodynamic equilibrium with radiation, and is given by

$$u = \frac{h\nu}{k_BT} = \frac{hc}{\lambda k_BT}. \tag{6.6}$$

We can equivalently write the Planck function in terms of wavelength as

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda k_BT}-1} = \frac{2k_B^5T^5}{h^4c^3}\frac{u^5}{e^u-1}. \tag{6.7}$$

Note that Equation (6.5) is *not* equivalent to Equation (6.7) when written equivalently using u rather than ν or λ. This implies that the shape and peak of the blackbody curve is dependent on whether it is written in wavelength or frequency space. Thus, the peak of the blackbody curve has no intrinsic physical meaning, as it depends on the units – see more below in Section 6.2.1.

Example blackbody radiance curves are plotted in Figure 6.3, over the same range of wavelength and frequency. Note first that cooler curves have smaller peak frequencies and

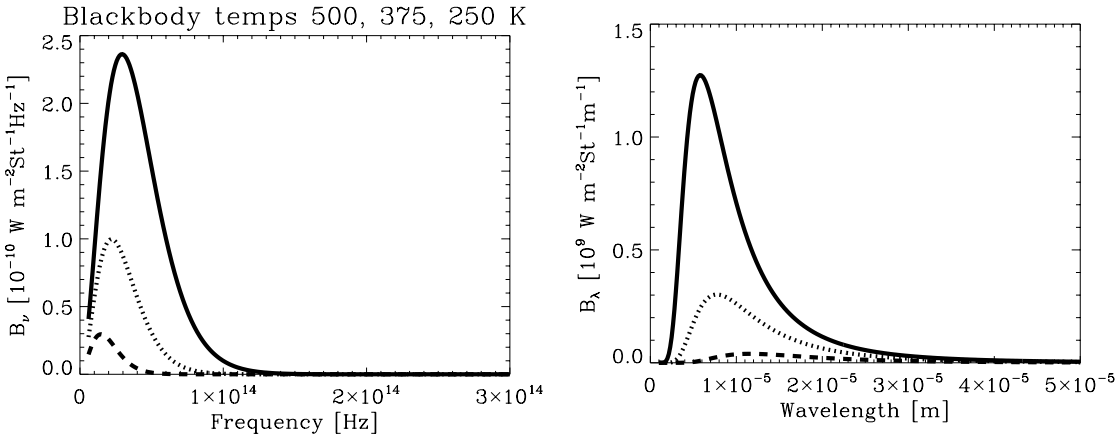


Figure 6.3: Blackbody curves for varying temperature emitters. Note how curves for different temperatures do not cross, and the shape (and peak) of the blackbody curves differs in frequency vs. wavelength space.

longer peak wavelengths. Additionally, cooler curves have smaller radiances, with curves for different temperatures never crossing. Finally, note that blackbodies of different temperature all peak at the same value of u.

We can relate the blackbody radiance B (which is the radiation energy per second meter squared per steradian of solid angle per unit of frequency or wavelength) to the energy flux F (which is the radiation energy per second per meter squared per unit of frequency or wavelength) by taking $F = \pi B$, that is

$$F_\nu(T) = \frac{2\pi h\nu^3}{c^2}\frac{1}{e^{h\nu/kT-1}} = \frac{2\pi k_B^3 T^3}{h^2c^2}\frac{u^3}{e^u-1}, \tag{6.8}$$

and

$$F_\lambda(T) = \frac{2\pi hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda k_B T}-1} = \frac{2\pi k_B^5 T^5}{h^4c^3}\frac{u^5}{e^u-1}. \tag{6.9}$$

6.2.1 Wien's law

The peak of the blackbody curve in frequency or wavelength space (i.e., the frequency where $\partial B/\partial\nu = 0$ or the wavelength where $\partial B/\partial\lambda = 0$) is given by Wien's law:

$$\nu_{\max} = 5.87\times10^{10}\frac{T}{\mathrm{K}}\ \mathrm{Hz}, \lambda_{\max}\ = 2897\frac{\mathrm{K}}{T}\ \mu\mathrm{m}. \tag{6.10}$$

An alternate way to characterise blackbody radiation more consistently is not by the peak frequency or wavelength, but by the *median emission frequency* that splits the energy flux such that half of it is contained at greater frequencies and half at lesser frequencies. This quantity does not depend on units, and so the median emission frequency or wavelength will occur at the same value of $u = 3.503$, intermediate between the peaks of B_ν and B_λ.

6.2.2 Stefan-Boltzmann law

If one integrates F_ν or F_λ over all frequencies or all wavelengths, respectively, one obtains the total flux emitted by a blackbody, known as the Stefan-Boltzmann law:

$$F(T) = \int_0^\infty \pi B_\nu(T)d\nu = \int_0^\infty \pi B_\lambda(T)d\lambda = \left[\frac{2\pi k_B^4}{h^3c^2}\int_0^\infty \frac{u^3}{e^u-1}du\right]T^4 = \sigma T^4. \tag{6.11}$$

In the equation above, $\sigma = 5.67\times10^{-8}$ W m^{-2} K^{-4} is the Stefan-Boltzmann constant. Note that the Stefan-Boltzmann constant only depends on two constants determining the non-classical nature of the universe (h and c) through the combination h^3c^2 – thus, in principle these two constants could trade off to provide a classical universe ($h\to0$, $c\to\infty$) while h^3c^2 stays fixed, allowing well-behaved blackbody radiation[28].

6.3 Simple energy balance models

We can use blackbody radiation to develop very idealised energy balance models to estimate planetary surface temperature, both ignoring and including the infrared radiative effect of an overlying atmosphere.

6.3.1 Equilibrium temperature

We previously derived the equilibrium temperature of Earth in our activity from Day 1. More generally, we can write a statement of energy balance for an isothermal object with no internal heat source as the power emitted equal to the power absorbed, or

$$f\pi R_p^2\epsilon\sigma T_{\mathrm{eq}}^4 = (1-\alpha)F_{\mathrm{inc}}\pi R_p^2, \tag{6.12}$$

[28]This was noted by Ray in his 2024 slides.

where R_p is the radius of the planet (or other object), f is a factor regarding how well heat is redistributed across the planet ($f = 4$ in the case of full redistribution across the sphere and $f = 1$ in the case of no redistribution whatsoever), ϵ is the emissivity, σ is the Stefan-Boltzmann constant, $T_{\rm eq}$ is the equilibrium emitting temperature, α is the albedo, and $F_{\rm inc}$ is the incoming (stellar) flux. We can thus solve for the equilibrium temperature as

$$T_{\rm eq} = \left[\frac{(1-\alpha)F_{\rm inc}}{f\epsilon\sigma}\right]^{1/4} . \tag{6.13}$$

Note that in the case of $f = 4$ one typically expresses $T_{\rm eq}$ as the "full-redistribution equilibrium temperature," while in the case of $f = 1$ one titles $T_{\rm eq}$ as the "irradiation temperature" or "zero-redistribution equilibrium temperature."

Using the inverse square law, we can express

$$F_{\rm inc} = F_\star = \frac{L_\star}{4\pi r_{\rm orb}^2}, \tag{6.14}$$

where $L_\star = 4\pi R_\star^2 \sigma T_\star^4$ from Stefan-Boltzmann and $r_{\rm orb}$ is the planet-star separation. Thus, for mature planets that have cooled off from formation (or more generally, planets that receive much more incident stellar power than their intrinsic cooling luminosity), one can approximate the planetary temperature by the equilibrium temperature (assuming here $\epsilon = 1$):

$$T_{\rm eq} = T_\star \left[\frac{(1-\alpha)}{f}\right]^{1/4} \sqrt{\frac{R_\star}{r_{\rm orb}}} . \tag{6.15}$$

In Equation (6.15), α is the Bond albedo, which is the fraction of total energy incident on the planet that is not absorbed and re-radiated (i.e., the amount that is scattered/reflected by clouds, haze, ice, and gas).

6.3.2 One-layer greenhouse & two-layer atmosphere

Please get in groups of two or three to work on the following activity deriving the surface temperature for a one-layer atmosphere, alongside the upper-atmospheric temperature in thermal energy balance.

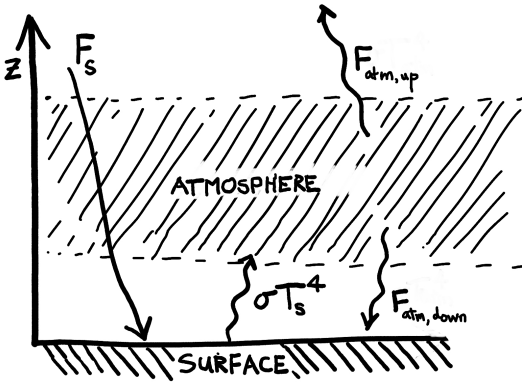


Figure 6.4: Schematic of the one-layer atmospheric greenhouse problem, courtesy Adam Showman.

1. Consider the one-layer atmosphere model shown in Figure 6.4. Assume that the atmosphere is completely transparent to shortwave radiation, and completely opaque to longwave radiation. Write down an expression for the surface energy balance, and solve it to find the surface temperature as a function of F_s and $F_{\text{atm,down}}$. How does this compare to the case without an atmosphere (i.e., the equilibrium temperature discussed above)?

2. Let us now consider what happens at very low pressures, where the gas becomes nearly transparent to the upwelling infrared radiation from below with $\epsilon_{\text{IR}} \ll 1$. Again neglect shortwave absorption in this region, so that the energy balance in this low pressure region is still between absorption and emission of IR. Given that the flux of IR upwelling radiation from the upper troposphere (below the nearly transparent low pressure region) is σT_e^4, write down another energy balance expression for the low-pressure region – to do so you should assume that the fraction of radiation absorbed (the absorptivity) is equal to the fraction emitted (the emissivity) – this is Kirchoff's law. Solve this to find the temperature of this low pressure region, which is termed the "skin temperature" for reasons we will discuss.

3. Now that we have simple models for the troposphere (where the greenhouse effect is strong) and the stratosphere (where the atmosphere converges to the skin temperature), put the two pieces together to develop a two-layer model of the temperature profile in a planetary atmosphere. Assume that the lapse rate in the troposphere is $dT/dz = \Gamma$.